DGL-26 Kaggle Competition: *Big Dawgs*, *Final Rank 4*

Vivian Lopez, 02196719, vl522

Saurajit Seth, 02314442, ss3722

Shivam Subudhi, 02207276, ss4522

Dhruv Himatsingka, 02217050, dh722

I. Methodology & Novelty (40 points)

A. Problem description & motivation (5 points)

Graph super-resolution aims to reconstruct a high-resolution (HR) graph from a low-resolution (LR) graph representing the same underlying structure. Formally, the task is to learn a mapping from a low-resolution adjacency matrix to its corresponding high-resolution adjacency matrix, enabling recovery of finer connectivity patterns.

In brain connectivity analysis, graphs model interactions between brain regions, where nodes represent regions of interest and edges represent functional or structural connections. LR graphs capture coarse interactions between larger regions, while HR graphs provide detailed connectivity by subdividing the brain into finer regions. Graph super-resolution seeks to infer detailed brain networks from coarser observations.

This problem is important because HR brain imaging data is expensive and noisy, whereas low-resolution data is easier to obtain. Effective super-resolution methods could enable detailed brain connectivity analysis without requiring costly data acquisition. Applications include identifying biomarkers for neurological disorders and improving downstream tasks such as brain network classification.

However, the task is challenging. The LR–HR mapping is ill-posed, as multiple HR graphs may correspond to the same LR graph [1]. Additionally, brain graphs are noisy and subject-specific, making generalisation difficult. Models must also increase resolution while preserving important topological properties, and must handle the dimensional mismatch between LR and HR graphs [2].

B. State-of-the art methods (3 points)

We have described SOTA models in Table I.

C. Main figure (4 points)

We have described the NeuroRes-GNN architecture in Figure 2.

TABLE I
Overview of GNN-based SOTA Models

Model Name	Brief Description
Cross-SRGANs [3]	Uses a cross-modal super-resolution generative adversarial framework that reconstructs higher-resolution brain representations from multimodal neuroimaging data (MRI and PET). The model learns inter-modal relationships through mutual learning and adversarial training, and the extracted features are processed by a GNN for disease classification.
CogGNN [4]	Introduces a generative GNN architecture that integrates structural brain connectivity with cognitive features. The model learns connectional brain templates using graph neural networks combined with a cognition-aware loss to generate biologically meaningful high-resolution brain networks.
Bi-SR & DEFEND [5]	Proposes two GNN-agnostic frameworks for graph super-resolution: Bi-SR, which uses a bipartite graph connecting LR and HR nodes to perform structure-aware node upsampling, and DEFEND, which maps HR edges to nodes in a dual graph to enable edge prediction via standard GNN layers while preserving graph topology.

D. Brief overview of the proposed GNN (5 points)

Figure 2 illustrates the end-to-end architecture of the proposed DenseGCN v3r model for brain graph super-resolution. The model receives a low-resolution (LR) brain connectivity graph $A_{LR} \in \mathbb{R}^{160\times160}$, where nodes correspond to brain regions and edges represent connectivity strengths. Node features are constructed directly from adjacency rows so that each node encodes its connectivity pattern with the rest of the graph.

The encoder consists of three residual DenseGCN message-passing blocks that transform the LR graph into latent node embeddings $H_{enc} \in \mathbb{R}^{160\times192}$. Each block performs graph convolution followed by LayerNorm, ReLU activation, dropout ($p = 0.35$) and a feed-forward network with width multiplier 4. A learnable edge bias and residual connections help stabilise training and improve representation capacity.

The encoded node representations are then mapped to the high-resolution (HR) node space through a linear upsampling

layer that expands the graph from 160 LR nodes to 268 HR nodes, producing embeddings $H_{hr} \in \mathbb{R}^{268\times192}$.

To reconstruct the HR graph, a multi-head bilinear decoder (4 heads) computes pairwise interactions between node embeddings to predict the HR adjacency matrix $\hat{A}_{HR} \in \mathbb{R}^{268\times268}$. A lightweight HR refinement layer further improves the reconstruction quality and outputs are clamped to the range $[0, \infty)$ to ensure valid edge weights.

During training, the model predicts the residual connectivity $(y - \bar{y})$ relative to the population mean graph and is optimised using an L1 reconstruction loss with the AdamW optimiser. Data augmentation strategies including mixup-style interpolation ($\lambda \sim \text{Beta}(0.2, 0.2)$) on vectorised LR/HR graph pairs, Gaussian noise ($\sigma = 0.01$) and edge dropout ($p = 0.05$) are applied to improve generalisation.

Overall, this encoder-upsampling-decoder architecture learns a structured mapping from LR graphs to HR graphs while preserving important connectivity patterns and graph topology.

E. Innovative components (10 points)

Compared with existing GNN-based brain graph super-resolution models, the proposed framework differs in both training strategy and prediction formulation. Methods such as Cross-SRGANs and CogGNN rely on generative modelling or multimodal supervision to reconstruct high-resolution brain representations, while Bi-SR and DEFEND address super-resolution through topology-expansion mechanisms using auxiliary graph structures. In contrast, our approach directly learns a mapping from LR to HR connectivity using a DenseGCN architecture, with two task-specific modifications designed for the small-sample paired setting.

Overall, our model does not treat brain graph super-resolution as a standard graph reconstruction problem. Instead, we reformulate it as a regularised residual prediction problem trained with graph-space interpolation. This makes the model better suited to the small-sample, inductive and structurally ambiguous nature of the dataset.

TABLE II
INNOVATIVE COMPONENTS OF THE PROPOSED GNN FRAMEWORK

Novel Contribution	**Rationale**
Adjacency-space mixup-style augmentation for paired brain graph super-resolution	Convex interpolation is applied directly to paired LR–HR adjacency matrices to generate additional structurally valid training samples while preserving symmetry and edge-weight constraints. This improves sample efficiency without requiring generative or multimodal training. This is a sample-wise mixup-style augmentation, inspired by the graphon-based G-Mixup method of Han et al [6].
Mean-residual disentanglement for HR graph prediction	Instead of predicting the full HR graph directly, the model is trained to predict the residual relative to the population mean HR graph and the mean is added back at inference time. This decomposes the task into a shared structural prior and a subject-specific deviation, allowing the network to focus on the harder personalised component of reconstruction. This formulation aligns well with brain connectivity data, where coarse topology is largely consistent across subjects while fine-grained edge variations differ. Unlike existing the SOTA approaches which learn the full HR graph via generative or topology-expansion mechanisms, our approach explicitly incorporates a population-level structural prior to regularise reconstruction.

F. Mathematical properties of the proposed GNN (13 points)

Permutation invariance (5 points)

a) A graph-level function f is *permutation invariant* if relabelling the nodes does not change the output. Formally, for any permutation matrix P, adjacency matrix A and node-feature matrix X,

$$f(PAP^\top, PX) = f(A, X).$$

In other words, the prediction depends only on the graph structure and node features, not on the arbitrary ordering of the nodes. (1 point)

b) Our full model is *not* permutation invariant. Permutation invariance would require exactly the same output after relabelling the input nodes, but our super-resolution model assumes a fixed anatomical ordering of brain regions. Although the DenseGCN encoder uses shared message-passing operations across nodes, the overall architecture includes a learned upsampling map from 160 LR nodes to 268 HR nodes and outputs HR edges in a fixed canonical order. Both steps depend on specific node identities, so relabelling the input nodes generally changes the prediction. Therefore, the model learns a mapping between graphs in a fixed atlas-based ordering rather than a permutation-invariant graph-level function. (4 points)

Permutation equivariance (5 points)

a) A graph-to-graph function f (which outputs an adjacency matrix) is *permutation equivariant* if permuting the input node ordering causes the output to be permuted in the same way. Formally, for any permutation matrix P,

$$f(PAP^\top, PX) = P\, f(A, X)\, P^\top.$$

(1 point)

b) Our model is permutation equivariant up to the learned upsampling stage. In the DenseGCN encoder, operations such as adjacency normalisation, message passing, shared linear transformations and residual connections are applied uni-

formly across nodes, so permuting the LR node ordering permutes the intermediate node embeddings in the same way. However, the subsequent upsampling layer from 160 LR nodes to 268 HR nodes is defined over fixed node indices, so it breaks exact permutation equivariance. The final HR prediction is also read out in a fixed upper-triangular edge order. Therefore, the encoder is equivariant up to upsampling, but the full end-to-end model is not fully permutation equivariant.(4 points)

Expressiveness (3 points)

a) Expressiveness refers to the ability of a GNN to represent and distinguish different graph structures, such as to capture sufficiently rich structural patterns for the task at hand. (1 point)

b) Our model is reasonably expressive. The DenseGCN encoder uses multiple layers of message passing, learnable transformations and nonlinear activations, so it can capture non-trivial structural patterns in the LR graph. Moreover, each node is represented by its full adjacency row, which provides rich connectivity information. However, because the model still relies on standard message passing, it is not maximally expressive in the strongest theoretical sense. (2 points)

II. Experimental Setup & Evaluation (27 points)

A. Results (9 points)

a) We introduce two additional topology-aware evaluation measures to complement the existing metrics.

TABLE III
Additional Topological/Geometric Measures

Measure Name	Brief Description & Rationale
Node Strength MAE	Mean Absolute Error of node strength (weighted degree). Measures how well the total incident edge weight per node is preserved between predicted and ground-truth HR graphs, capturing hubness and connectivity importance. We selected this metric because the existing measures mainly evaluate edge-level similarity (MAE, PCC, JSD) or global centrality scores. Node strength provides a simple and efficient measure of hub connectivity in weighted graphs, allowing us to assess whether highly connected regions maintain their overall connectivity intensity.
Clustering Coefficient MAE	Mean Absolute Error of node clustering coefficients. Evaluates preservation of local triangle density and community structure in the reconstructed graphs. Unlike the provided metrics, which focus on edge accuracy or node importance, the clustering coefficient captures mesoscale topology by measuring local community structure in the graph.

b) We evaluated our model using 3-fold cross-validation on the 167 training samples. Performance was measured using eight evaluation metrics: MAE, PCC, JSD, PageRank centrality MAE, Eigenvector centrality MAE, Betweenness centrality MAE, Strength MAE and Clustering coefficient MAE. Table IV summarises the results across the three folds.
The model achieved an average MAE of 0.140 and PCC of 0.614, with Jensen-Shannon distance of 0.291. Structural metrics also indicate strong topology preservation, including MAE(Node Strength) = 0.256 and MAE(Clustering) = 0.061. The small variation across folds demonstrates stable performance and good generalisation despite the relatively small dataset.

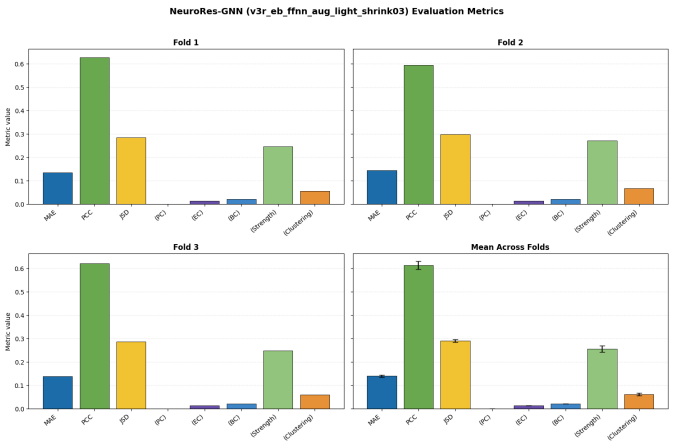

Fig. 1. Evaluation metrics across the three cross-validation folds and their mean.

TABLE IV
3-Fold Cross-Validation Performance Across Evaluation Metrics

Metric	Fold 1	Fold 2	Fold 3	Mean
MAE	0.1358	0.1454	0.1392	**0.1401**
PCC	0.6276	0.5946	0.6210	**0.6144**
JSD	0.2859	0.2979	0.2877	**0.2905**
MAE (PC)	0.000566	0.000571	0.000566	**0.000568**
MAE (EC)	0.013665	0.014093	0.013316	**0.01369**
MAE (BC)	0.020647	0.020943	0.020904	**0.02083**
MAE (Node Strength)	0.2474	0.2715	0.2485	**0.2558**
MAE (Clustering)	0.0569	0.0675	0.0599	**0.0614**

c) The full 3-fold cross-validation training procedure required 3468.8s (57.81 min) on a GPU-enabled environment (NVIDIA GeForce RTX 4080 Laptop GPU). Peak RAM usage during training was 1.887 GB.

d) For the Kaggle competition evaluation, the final model was retrained on the complete training dataset ($train_LR$ and $train_HR$) before generating predictions for the $test_LR$ samples. The predictions were post-processed by clipping negative values and applying mild shrinkage before serialisation into the required submission

format. Our best public submission achieved a Mean Absolute Error (MAE) of 0.127094, initially securing 6th place among 24 teams on the public leaderboard. **Upon transition to the private leaderboard, our model demonstrated superior generalisation, ascending to 5th place with a final MAE of 0.148109. Notably, this performance was highly competitive, falling within 0.74% of the first-place score (0.147022).**

B. Comparison Against Other Methods *(6 points)*

We compare our proposed NeuroRes-GNN model against two reference methods: (1) a naive Simple Graph Convolution (SGC) model and (2) a Variational Graph Autoencoder (VGAE), which serves as a representative state-of-the-art graph reconstruction approach. All models were evaluated using the same 3-fold cross-validation protocol on the 167 training samples to ensure identical train/validation splits and comparable experimental conditions.

a) Table V presents the average performance across the three folds for all models using the eight evaluation measures. These metrics evaluate both edge reconstruction accuracy and preservation of graph topology.

TABLE V
MODEL PERFORMANCE COMPARISON (3-FOLD CROSS-VALIDATION). BEST VALUES FOR EACH METRIC ARE HIGHLIGHTED IN BOLD.

Metric	SGC	VGAE	NeuroRes-GNN
MAE	0.1423	0.1422	**0.1401**
PCC	0.6174	**0.6181**	0.6144
JSD	0.2916	0.2915	**0.2905**
MAE (PC)	**0.000559**	**0.000559**	0.000568
MAE (EC)	**0.01367**	**0.01367**	0.01369
MAE (BC)	0.01948	**0.01935**	0.02083
MAE (Strength)	0.2606	0.2605	**0.2558**
MAE (Clustering)	0.0608	**0.0558**	0.0614

Overall, all three models achieve very similar performance across most metrics. NeuroRes-GNN achieves the lowest reconstruction error (MAE $= 0.140$) and slightly better preservation of node strength, while VGAE achieves the best clustering preservation and slightly higher PCC. These results suggest that the graph super-resolution task can be captured effectively even by relatively simple architectures. Nevertheless, our model achieves the best reconstruction accuracy and strength preservation, indicating modest benefits from the additional architectural components.

b) Our initial expectation was that NeuroRes-GNN would outperform both SGC and VGAE due to its more expressive architecture, which incorporates residual message passing, adjacency-row node features and a bilinear decoder with learnable edge bias. While NeuroRes-GNN does achieve the best MAE and strength preservation, the performance differences between models are relatively small.

One surprising observation is that the simpler SGC model performs competitively across most metrics. This may indicate that the mapping between LR and HR graphs is largely linear, allowing even shallow graph convolution models to capture the relationship effectively. In contrast, VGAE slightly improves clustering preservation, likely due to its latent embedding structure, which can capture local connectivity patterns.

C. Scalability of Your Proposed GNN Model *(7 points)*

We analyse the scalability of the proposed DenseGCN-based NeuroRes-GNN architecture with respect to both dataset size and graph resolution.

Computational complexity. The encoder consists of three GCN blocks operating on the LR graph with $n = 160$ nodes and hidden dimension $d = 192$. Each message-passing layer performs multiplications of the form SHW, where $S \in \mathbb{R}^{n\times n}$ is the normalised adjacency matrix, $H \in \mathbb{R}^{n\times d}$ the node embeddings, and $W \in \mathbb{R}^{d\times d}$ the learnable weights. This results in a computational complexity of $O(n^2d + nd^2)$ per layer. The HR reconstruction stage uses a bilinear decoder to produce the predicted adjacency matrix $\hat{A}_{HR} \in \mathbb{R}^{m\times m}$ with $m = 268$, requiring $O(md^2 + m^2d)$ operations, dominated by the $O(m^2d)$ term due to dense adjacency reconstruction. Consequently, the HR decoding stage constitutes the primary computational cost of the model.

Scaling with dataset size. Training cost increases approximately linearly with the number of samples because each graph is processed independently within mini-batches. In our experiments, performing full 3-fold cross-validation on 167 subjects required approximately 57.8 minutes of training time. Increasing the dataset size therefore primarily increases total training time, while GPU memory usage remains largely determined by the batch size and graph resolution. This makes the model practical for larger cohorts.

Memory usage. The model contains approximately 14 MB of parameters and requires roughly 50 MB of GPU memory during training with batch size 16. Memory consumption is primarily dominated by storing intermediate node embeddings and decoder activations during backpropagation, as well as dense HR adjacency predictions. Because graph resolution is fixed in our experiments, GPU memory usage remains largely constant as the number of samples in the dataset increases.

Scaling with graph resolution. The primary scalability limitation arises from the dense HR decoder. Since the predicted adjacency matrix is constructed explicitly, both computational cost and memory usage scale quadratically with the number of HR nodes. Specifically, the bilinear decoder has complexity $O(m^2d)$ for m HR nodes and embedding dimension d. For example, increasing the HR graph size from 268 to 500 nodes increases decoder computation by approximately $3.5\times$. While manageable for medium-sized brain atlases, substantially higher-resolution graphs would significantly increase both runtime and memory requirements.

Potential improvements. Scalability could be further improved by replacing the dense bilinear decoder with a low-rank reconstruction model or by adopting approximate sparse adjacency representations. Such approaches could reduce the

TABLE VI
ESTIMATED SCALING BEHAVIOR OF THE PROPOSED MODEL

Scenario	Training Time	Memory
Current dataset (167 subjects)	∼58 min	∼50 MB
Large cohort (1000 subjects)	∼5.8 h	∼50 MB
Higher resolution (500 nodes)	∼1.5 h	∼150 MB

$O(m^2 d)$ decoding cost while maintaining structural reconstruction quality.

Overall, the proposed architecture scales well for moderate dataset sizes and graph resolutions. However, for substantially larger HR node counts, more efficient decoding strategies such as sparse or low-rank reconstruction would be required to maintain computational efficiency.

D. Reproducibility of Your Proposed GNN Model (5 points)

The proposed DenseGCN-based model was designed to ensure both experimental reproducibility and stable performance across different train/test splits.

First, all sources of randomness are controlled using a fixed random seed (seed $= 42$). The seed is applied consistently to Python, NumPy and PyTorch random number generators, while CUDA operations are configured for deterministic execution. In addition, the 3-fold cross-validation procedure uses `KFold(shuffle=True, random_state=42)`, ensuring identical data splits across repeated runs.

Second, model configurations and hyperparameters are stored in predefined presets within the training script (`train_dense_gcn.py`). These presets define architectural and optimisation settings such as the number of GCN layers, hidden dimension, dropout rate, learning rate and optimiser. Model checkpoints are also saved during training, allowing experiments to be reproduced or resumed reliably.

In terms of empirical consistency, the model demonstrates stable performance across the three cross-validation folds. For example, the MAE values across folds were 0.136, 0.145 and 0.139, yielding a small standard deviation of approximately 0.004. Similar stability is observed for other metrics such as PCC and JSD, indicating that the model generalises consistently even when the training and validation distributions vary between folds.

This stability can be attributed to several architectural and training choices, including residual message passing, dropout regularisation and mixup-style augmentation, which help prevent overfitting and improve robustness to variations in the training data.

Finally, reproducibility is supported by explicit dependency management through a `requirements.txt` file specifying the software environment used during training. Together, these measures ensure that both the training procedure and the resulting model performance can be reproduced reliably across different runs and computing environments.

III. DISCUSSION & REFLECTIONS (8 POINTS)

a) We critically evaluated the strengths and limitations of our proposed DenseGCN-based NeuroRes-GNN model. Overall, the model demonstrates strong performance in reconstructing HR graphs while preserving important topological properties. However, several limitations related to architectural design and evaluation methodology were also identified.

TABLE VII
STRENGTHS & WEAKNESSES OF THE PROPOSED GNN

Strength 1	The model preserves important structural properties of the HR graphs. For example, the reconstructed graphs achieve low errors for topology-based metrics such as node strength (MAE $\approx$ 0.256) and clustering coefficient (MAE ≈ 0.061), indicating that the model captures both global connectivity patterns and local community structure.
Strength 2	The architecture incorporates several techniques that improve robustness on a relatively small dataset (167 samples). In particular, residual message passing, dropout regularisation and mixup-style augmentation help prevent overfitting and lead to stable performance across cross-validation folds.
Weakness 1	Performance differences between our model and simpler baselines were relatively small. For example, SGC achieved similar reconstruction accuracy across several metrics, suggesting that the LR-to-HR mapping may be partially captured by simpler models and limiting the observed benefit of the more complex architecture.
Weakness 2	The model relies on dense HR graph reconstruction through a bilinear decoder, which scales quadratically with the number of nodes. While manageable for the current graph size (268 nodes), this design could become computationally expensive for larger brain atlases or higher-resolution graphs.

b) Several improvements could further strengthen the proposed approach. First, incorporating additional topology-aware objectives, such as spectral alignment or Laplacian consistency losses, could encourage stronger preservation of global graph structure during training. Second, future work should perform a more systematic ablation study to quantify the contribution of architectural components such as mixup-style augmentation, residual connections and edge bias in the decoder. Understanding which components contribute most to performance could guide the design of more efficient graph super-resolution models.

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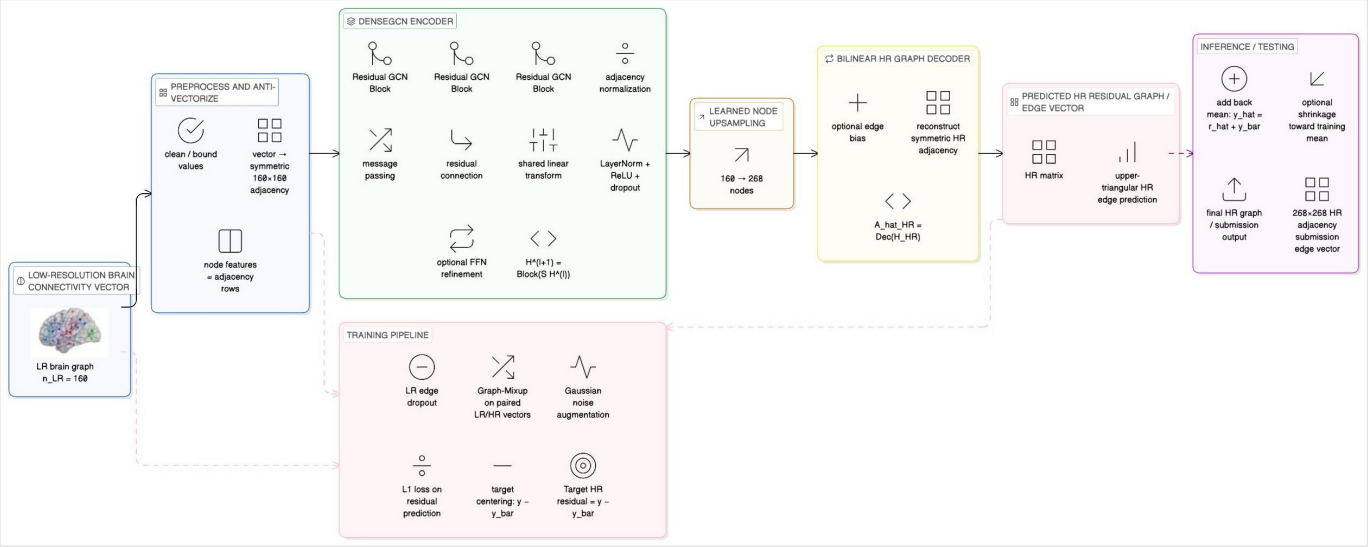

Fig. 2. Proposed architecture of NeuroRes-GNN

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